MILITARY SPECIFICATION

CONNECTOR, HOSE, QUICK-ACTING, MALE, ANTI-G GARMENT

1. SCOPE

1.1 This specification covers one type of quick-acting, male connector for the hose of an anti-g garment. The connector is designated MXU-636/P.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Federal

QQ-S-764	Steel Bar, Corrosion Resisting, Free Machining

Military

MIL-H-775	Hose, Rubber, Plastic Fabric or Metal (Including Tubing); and Fittings, Nozzle and Strainers, Packaging of
MIL-H-6875	Heat Treatment of Steels (Aircraft Practices, Process for)
MIL-H-26385	Hose, Oxygen and Pressurization, Ozone Resistant

STANDARDS

Military

MIL-STD-100	Engineering Drawing Practices
MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-130	Identification Marking of US Military Property
MIL-STD-143	Standards and Specifications, Order of Precedence for the Selection of
MIL-STD-831	Test Reports, Preparation of
MIL-STD-889	Dissimilar Metals
MS27755	Connector, Hose Quick Disconnect, Female, Anti "G" Suit

DRAWING

Air Force

55B3674	Hose, Quick Disconnect Anti-G Suit

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(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publication. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

STANDARD

American National Standards Institute (ANSI)

ANSI (formerly ASA) B46.1 Surface Texture

(Copies of this standard may be obtained from the American National Standards Institute, 10 East 40th Street, New York, New York 10016.)

(Technical society and technical association specifications and standards are generally available for reference from libraries and are also distributed among technical groups and using federal agencies.)

3. REQUIREMENTS

3.1 Preproduction. This specification provides for preproduction testing (see 4.4 and 6.2).

3.2 Selection of specifications and standards. Specifications and standards for necessary commodities and services not specified herein shall be selected in accordance with MIL-STD-143.

3.3 Materials. Materials shall conform to applicable specifications and standards and shall be as specified herein. Materials that are not covered by specifications and that are not specifically described herein shall be the best quality and the lightest practicable weight and shall have been tested to assure suitability for the intended use.

3.3.1 Metals. Metals not specified herein shall be corrosion resistant or suitably treated to resist corrosion due to fuels, salt spray, or atmospheric conditions likely to be met in storage or normal use. The metal used to fabricate the nose piece of the connector shall be corrosion-resistant steel conforming to type 416, condition A of QQ-S-764. All metal surfaces shall be finished to 125AA in accordance with ANSI (formerly ASA) B46.1. The nose piece shall be heat treated to Rockwell hardness C39 - C41 in accordance with MIL-H-6875. All burrs shall be removed from the metals.

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3.3.1.1 Dissimilar metals. Dissimilar metals shall not be used in contact with each other. Dissimilar metals are defined in MIL-STD-889.

3.3.2 Protective treatment. Materials that will be subject to deterioration during service usage shall be protected against deterioration in a way that will not affect compliance with the performance requirements specified herein. Protective coatings that will crack, chip, or scale with age or extremes of environmental conditions shall not be used.

3.4 Design and construction. The connector shall be designed and constructed, in accordance with figure 1, to be installed on an anti-g garment to provide a quick connection and a quick disconnection of the anti-g garment to and from an aircraft-installed pressure source. When the connector is joined to a female connector conforming to MS27755, a plunger shall open a water check valve and allow free flow of air to and from the anti-g garment. After the connectors are disconnected from each other, the water check valve shall close and prevent water from entering the anti-g garment while submerged in water. The connector shall also have a pressure bleed provision that will insure excess pressure will not remain in the anti-g garment when the connector is separated from the pressure source in an emergency.

3.5 Performance

3.5.1 Inflation and deflation. The connector shall not restrict normal inflation of the anti-g garment to 8 pounds per square inch (psi) within 3 seconds. The connector shall not restrict deflation of the anti-g garment within 4 seconds while the anti-g garment is connected to a female connector conforming to MS27755.

3.5.2 Emergency pressure relief. When an anti-g garment, pressurized to 8 psi, is disconnected from the air pressure source, the connector shall provide for deflation of the anti-g garment to 1.5 psi internal pressure within 4 seconds and shall provide for continued deflation of the anti-g garment to 0.5 (plus 0.25 or minus 0) psi. The anti-g garment shall maintain the internal pressure of not less than 0.5 psi for a 4-hour period.

3.5.3 Separation force. The force required to separate the connector from a female connector conforming to MS27755 shall be not less than 5 pounds nor more than 20 pounds.

3.5.4 Air leakage. The connector shall not leak air and shall not separate from the female connector when the joined connectors are submerged in water and subjected to an internal air pressure of 40 psi for 60 seconds.

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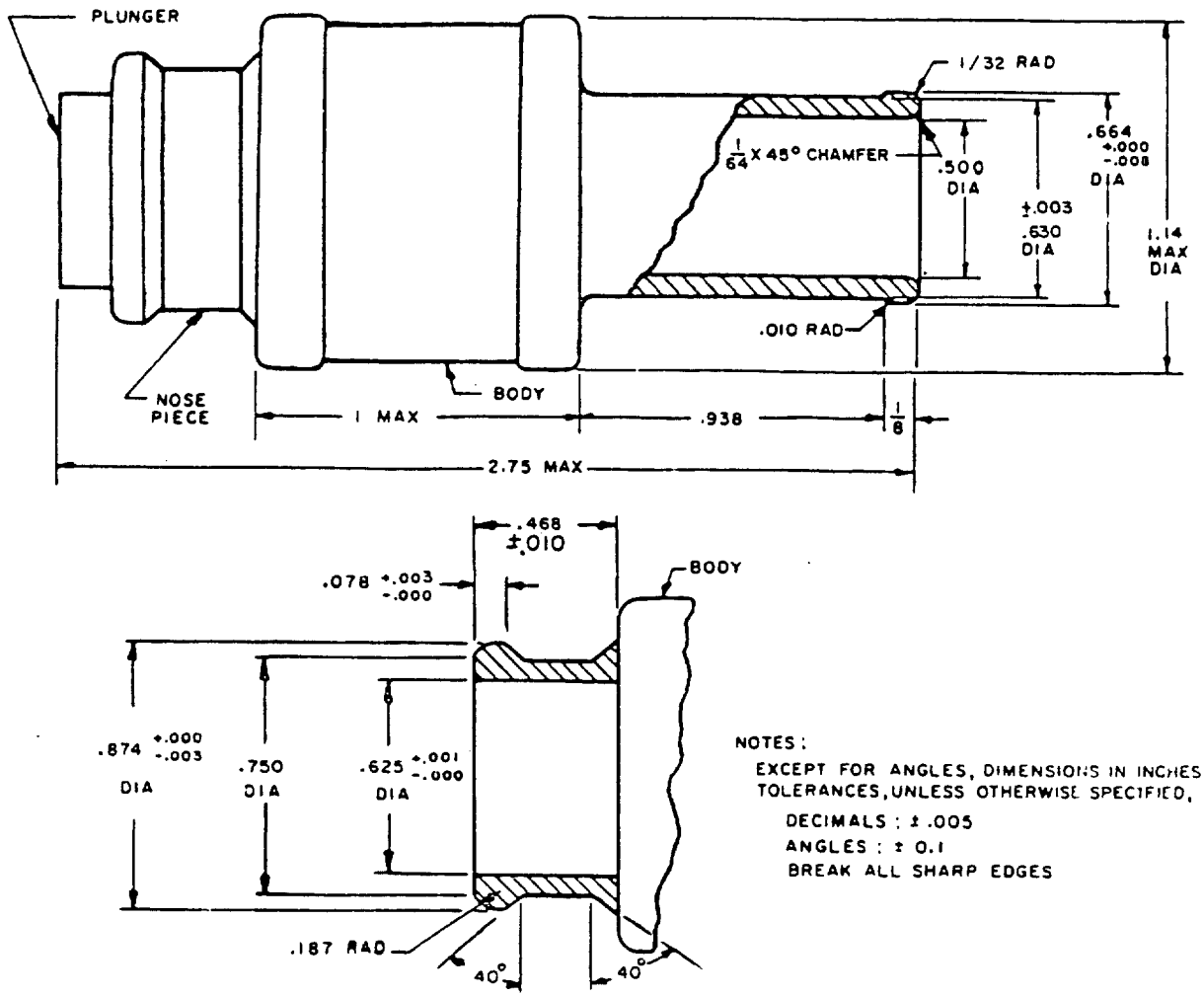


FIGURE 1. Connector

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3.5.5 Environmental exposure

3.5.5.1 High temperature. The connector shall function satisfactorily after exposure to a temperature of plus 160 (plus or minus 5) degrees Fahrenheit for a 4-hour period.

3.5.5.2 Low temperature. The connector shall function satisfactorily after exposure to a temperature of minus 60 (plus or minus 5) degrees Fahrenheit for a 4-hour period.

3.5.6 Reliability. The connector shall have a minimum mean cycles between failures of 300 cycles with a minimum reliability of 0.90 at a confidence factor of 0.90.

3.6 Identification of product. The connector shall be marked for identification in accordance with MIL-STD-130.

3.7 Government-loaned property. When requested by the contractor (see 4.3.2 and 6.2), a connector conforming to MS27755 and a CSU-12/P anti-g garment in size large to be used for test purposes will be loaned by the procuring activity.

3.8 Interchangeability. All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable. The item identification and part number requirements of MIL-STD-100 shall govern the manufacturer's part numbers and changes thereto.

3.9 Workmanship. The connector shall be free of defects, burrs, sharp edges, rough grinding marks, and uneven surfaces due to forming operations. Particular attention shall be given to accuracy of dimensions; radii of fillets; marking of parts and assemblies; thoroughness of welding, brazing, painting, swaging, soldering, and potting; alignment of parts; and tightness of assembly screws and bolts. Operation of the connector after assembly shall not be dependent upon selective fit of any or all parts.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.2 Classification of inspection. The examinations and the tests required by this specification are classified as follows:

(a) Preproduction tests

(b) Quality conformance tests.

4.3 Test conditions

4.3.1 Atmospheric conditions. Unless otherwise specified herein, all tests required by this specification shall be made at an atmospheric pressure of 30 (plus or minus 2) inches of mercury, at a temperature of plus 77 (plus or minus 10 degrees) Fahrenheit, and at a relative humidity of 80 percent or less. Where tests are made with atmospheric pressure or temperature substantially different from these values, proper allowance shall be made for the change in instrument readings.

4.3.2 Test equipment. The female connector required for the tests specified herein shall conform to MS27755. The anti-g garment required for the tests specified herein shall be a CSU-12/P anti-g garment in size large.

4.4 Preproduction tests. The preproduction tests shall consist of all tests described under 4.6. The preproduction samples shall be subjected to the preproduction tests. The preproduction samples shall consist of two connectors that are representative of the materials, the components, the construction, and the workmanship to be used during the production of the connectors. After the preproduction tests have been completed, the tested preproduction samples shall be submitted to the procuring activity (see 6.2). When specified on DD Form 1423 and by the procuring activity (see 6.2), one copy of a preproduction test report, in the format specified in MIL-STD-831, shall be submitted with the tested preproduction samples. Production of the connectors shall not be commenced prior to the approval of the preproduction samples by the procuring activity. Approval or waiver of the preproduction samples shall not preclude the requirements for the quality conformance tests.

4.5 Quality conformance tests. The quality conformance tests shall consist of the following:

(a) Individual test

(b) Sampling tests.

4.5.1 Individual test. Each connector shall be subjected to the examination of materials and workmanship described in 4.6.1.1.

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4.5.2 Sampling tests. Samples shall be selected in accordance with MIL-STD-105 and subjected to the sampling tests listed in table I and described herein. The sample unit for the sampling tests shall be one connector. The number of sample units (sample size) and the acceptance criteria for the sampling tests shall be as specified in table I.

TABLE I. Sampling Tests

Sampling Test and Paragraph Reference	Inspection Level of MIL-STD-105 for Selection of Sample Size 1/	Accept-Reject Criteria
Examination of marking (see 4.6.1.2)	II	2/
Examination of dimensions (see 4.6.1.3)	S-1	Ac 0 R_e 1
Inflation and deflation (see 4.6.2)	S-1	Ac 0 R_e 1
Emergency pressure relief (see 4.6.3)	S-1	Ac 0 R_e 1
Separation force (see 4.6.4)	S-1	Ac 0 R_e 1
Air leakage (see 4.6.5)	S-1	Ac 0 R_e 1
Environmental exposure (see 4.6.6.1 and 4.6.6.2)	S-1	Ac 0 R_e 1
Reliability (see 4.6.7)	S-1	Ac 0 R_e 1

1/ The sample size for inspection level S-1 of MIL-STD-105 shall be based only on the applicable sample size code letter corresponding to the specified inspection level of MIL-STD-105.
2/ The acceptable quality level for minor defects (examination of marking) shall be 2.5 defects per 100 units.

4.5.3 Defects in items already accepted. The investigation of a test failure could indicate that defects may exist in items already accepted. If so, the contractor shall fully advise the procuring activity, in writing, of all the defects likely to be found and the methods of correcting them.

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4.6 Test methods

4.6.1 Examination of product

4.6.1.1 Examination of materials and workmanship. The connector shall be examined, for the critical defects listed in table II, to determine compliance with the requirements of this specification with respect to materials and workmanship. If the connector contains any of the critical defects, the connector shall be rejected.

TABLE II. Classification of Defects

No.	Defect	Critical	Minor
1	Foreign matter embedded in materials	X	
2	Surface unclean, rough, or misaligned, or surface containing cracks, nicks, or any other flaws	X	
3	Any component missing, malformed, fractured, or otherwise damaged	X	
4	Any component loose or otherwise not securely retained	X	
5	Any component incorrectly assembled or improperly positioned	X	
6	Any functioning part that works with difficulty	X	
7	Faulty workmanship or any other irregularities	X	
201	Marking missing, insufficient, incorrect, illegible, or not permanent		X

4.6.1.2 Examination of marking. The connector shall be examined, for the minor defect listed in table II, to determine compliance with the marking requirements of this specification.

4.6.1.3 Examination of dimensions. The connector shall be examined to determine if the dimensions of the connector are as specified in figure 1. Each dimension of the connector that is not within the specified tolerance shall be classified as a defect (one class).

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4.6.2 Inflation and deflation. The connector shall be installed on a hose that conforms to Drawing 55B3674. The hose shall be attached to an anti-g garment or a 10-liter test tank. The connector shall then be attached to a female connector that conforms to MS27755. The female connector shall be connected to a source of air that has a free flow of 10 cubic feet per minute and a means to adjust the pressure to a maximum 12 psi pressure. The pressure shall be adjusted so that the anti-g garment or the tank will be pressurized to 8 psi within 3 seconds. The air source between the regulator and the female connector shall be disconnected. The anti-g garment or the tank shall then be permitted to bleed down to zero gage pressure. Restriction to pressure bleed down shall be noted. Failure of the connector during this test to meet any of the requirements specified in 3.5.1 shall be cause for rejection of the connector.

4.6.3 Emergency pressure relief. The connector shall be installed on a hose that conforms to Drawing 55B3674. The hose shall be attached to an anti-g garment or to a 10-liter test tank. The connector shall then be attached to a female connector that conforms to MS27755. The female connector shall then be connected to a source of air with a maximum pressure of 12 psi. The anti-g garment or the tank shall be pressurized to 8 psi, and the connectors shall be separated from each other. The pressure in the anti-g garment or the tank and the times during pressure bleed down shall be recorded. Failure of the connector during this test to meet any of the requirements specified in 3.5.2 shall be cause for rejection of the connector.

4.6.4 Separation force. Two 10-inch-long pieces of 5/8-inch-inside-diameter hose conforming to MIL-H-26385, or equivalent, shall be used for this test. An end of one of the pieces of hose shall be clamped to a female connector conforming to MS27755. An end of the other piece of hose shall be clamped to the male connector covered by this specification. The free end of the hose attached to the female connector shall be rigidly secured so that the hose and the female connector will be suspended vertically. The male connector shall then be connected to the female connector. A vertical downward force shall be applied to the free end of the hose that is attached to the male connector. The initial magnitude of this force shall be 4 pounds or less and shall be slowly increased until the connectors are disconnected. The magnitude of the force at disconnection shall be recorded and shall be within the limits specified in 3.5.3.

4.6.5 Air leakage. The connector shall be joined to a female connector. The free end of the female connector shall be temporarily sealed. The free end of the other connector shall be connected to an air source that has a pressure of at least 40 psi. A device that can measure air pressure shall be installed so that the pressure applied to the connectors can be recorded. The joined connectors shall be submerged in water, and an air pressure of 40 psi shall be applied for a 60-second period. While submerged the connectors shall be examined for leakage. Ascending air bubbles shall be evidence of leakage. Any leakage or separation shall be cause for rejection of the connector.

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4.6.6 Environmental exposure

4.6.6.1 High temperature. The connector shall be placed within a chamber and heated to a temperature of plus 160 (plus or minus 5) degrees Fahrenheit. The connector shall be maintained at this temperature for a 4-hour period. The connector shall then be allowed to cool to room temperature and shall be subjected to and shall pass the tests specified in 4.6.2 and 4.6.4.

4.6.6.2 Low temperature. The connector shall be placed in a chamber and cooled to and maintained at a temperature of minus 60 (plus or minus 5) degrees Fahrenheit for a 4-hour period. At the end of the 4-hour period, the connector shall be permitted to return to room temperature and then shall be subjected to and shall pass the tests specified in 4.6.2 and 4.6.4.

4.6.7 Reliability. The connector shall be subjected to the environmental exposure tests specified in 4.6.6.1 and 4.6.6 2, in sequence, except that the connector shall not be subjected to the tests specified in 4.6.2 and 4.6.4 after each environmental exposure test. After the connector has been subjected to the two exposures, the connector shall be subjected to 700 cycles of connection to and disconnection from a female connector conforming to MS27755. A cycle shall consist of a complete connection and disconnection. The connection shall consist of the following: The connectors shall be firmly joined. The internal water check valve shall be open, permitting free flow of air through the joined connectors. Disconnection shall consist of a complete separation of the connectors from each other. After the 700th cycle, the connector shall be subjected to and shall pass the tests specified in 4.6.2 through 4.6.5, to determine if failure has occurred. A connector failing to pass any of the tests specified herein shall have failed the reliability test. No parts shall be replaced during reliability testing.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be level A or C of MIL-H-775 as specified (see 6.2).

5.2 Packing. Packing shall be level A, B, or C of MIL-H-775 as specified (see 6.2).

6. NOTES

6.1 Intended use. The connector covered by this specification is intended to be installed on an anti-g garment and to be used with a female connector that conforms to MS27755 to provide a connection between the hose of the anti-g garment and an aircraft-installed air pressure source.

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6.2 Ordering data. Procurement documents should specify the following:

(a) Requirements for preproduction (see 3.1 and 4.4)

(b) Government-loaned property (see 3.7 and 4.3.2)

(c) Selection of applicable levels of preservation and packaging and packing (see 5.1 and 5.2).

Custodian:
Air Force - 11

Reviewer:
Air Force - 82

Preparing activity:
Air Force - 11

Project No. 4730-F008

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

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DOCUMENT IDENTIFIER AND TITLE

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

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